Science of the Total Environment 634 (2018) 1615–1630

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Biogeochemistry of natural ponds in agricultural landscape: Lessons learned from modeling a kettle hole in Northeast Germany

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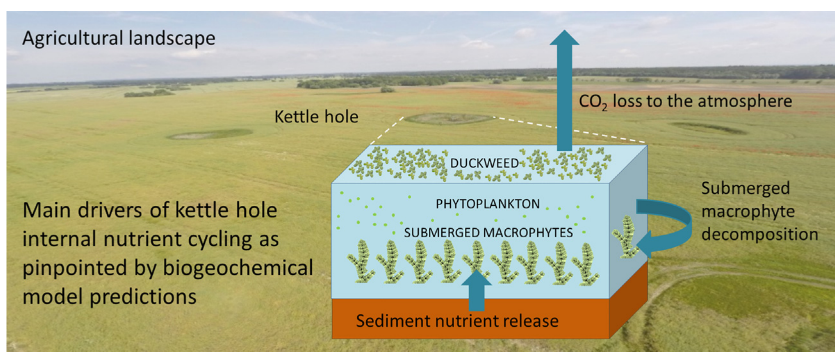
HIGHLIGHTS

- A biogeochemical model was developed for a kettle hole in an agricultural landscape.
- The model predicts nutrient and primary producer seasonal dynamics.
- Low-light tolerance and differential nutrient use shape the autotrophic assemblage.
- Sediment nutrient release and submerged macrophyte decay drive nutrient cycling.
- The pond acts as a net source of CO_2 to the atmosphere on an annual scale.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 9 February 2018
Received in revised form 30 March 2018
Accepted 1 April 2018
Available online xxxx

Editor: D. Barcelo

Keywords:
Kettle holes
Temporary ponds
Macrophytes
Phytoplankton
Water quality
Carbon flux
Internal nutrient recycling

ABSTRACT

Kettle holes, small shallow ponds of glacial origin, represent hotspots for biodiversity and biogeochemical cycling. They abound in the young moraine landscape of Northeast Germany, potentially modulating element fluxes in a region where intensive agriculture prevails. The Rittgarten kettle hole, with semi-permanent hydroperiod and a surrounding reed belt, can be considered as a representative case study for such systems. Aiming to provide insights into the biogeochemical processes driving nutrient and primary producer dynamics in the Rittgarten kettle hole, we developed a mechanistic model that simulates the carbon, nitrogen, phosphorus and oxygen, phytoplankton, and free-floating macrophyte biomass dynamics. After model calibration and sensitivity analysis, our modeling exercise quantified the simulated nutrient fluxes associated with all the major biogeochemical processes considered by the model. Seasonality of nutrient concentrations, magnitude of primary productivity rates, and biogeochemical process characterization in the pond were reasonably reproduced by the model from July 2013 to July 2014. Our results suggest that the establishment of a phytoplankton community well-adapted to low light availability, together with the differential use of N and P from free-floating macrophytes and phytoplankton can explain their coexistence in kettle holes. Sediment nutrient release along with decomposition of decaying submerged macrophyte are essential drivers of internal nutrient cycling in kettle holes. Our results also suggest that the Rittgarten kettle hole act as a net source of CO_2 to the atmosphere on an annual scale, which offers a testable hypothesis for kettle holes with structural and functional similarities. We conclude by

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https://doi.org/10.1016/j.scitotenv.2018.04.014

discussing the need to shed light on the effects of water level fluctuations on nutrient dynamics and biological succession patterns, as well as the relative importance of external sources and internal nutrient recycling mechanisms.

1. Introduction

Small lentic water bodies (<1 ha) dominate inland waters, both in terms of area and numbers (Downing et al., 2006). Their areal contribution has been underestimated in former times and so has been their role in global cycles (Downing, 2010). Recently, small inland waters such as kettle holes have been increasingly recognized as hotspots of carbon (C) turnover in the landscape (Aufdenkampe et al., 2011; Cole et al., 2007; Premke et al., 2016; Tranvik et al. 2009). Nevertheless, most of the research on C and nutrient cycling is conducted either in large lakes or in shallow lakes or artificial small standing inland waters, whereas the biogeochemistry and ecology of kettle holes have received less attention. Kettle holes are small (< 1 ha) shallow ponds that mostly developed in glacial depressions in extensive areas of North America and Europe when dead ice blocks melted during the retreat of the ice at the end of the last glaciation (Detenbeck et al., 2002; Lischeid and Kalettka, 2012; Tiner, 2003). In Northeast Germany, a region where intensive agriculture prevails, kettle holes are predominant landscape features. They amount to a total number of 150,000 to 300,000 and a density up to 40 km^{-2}, covering as much as 5% of arable land (Kalettka and Rudat, 2006).

Kettle holes vary widely with respect to hydrogeomorphology, dominant vegetation, hydroperiod and water quality (Kalettka and Rudat, 2006; Lischeid and Kalettka, 2012). The temporal and spatial variability in the aforementioned properties confers them the potential to be hotspots for biogeochemical cycling and biodiversity (Altenfelder et al., 2014; Gołdyn et al., 2012; Lischeid and Kalettka, 2012; Pätzig et al., 2012, Reverey et al., 2016). However, knowledge on the factors controlling water quality, including both external drivers and internal processes, is limited (Gołdyn et al., 2015; Lischeid et al., 2017). There is only little information on the effects of drying and rewetting on nutrient turnover and microbiology in these temporary systems (Attermeyer et al., 2017; Reverey et al., 2016). Similarly, knowledge on their biogeochemistry and contribution to global nutrient cycles is limited (Lischeid et al., 2017; Reverey et al., 2016). Overall, when compared to other inland water systems, nutrient cycling in small inland waters as ponds, and particularly kettle holes, remains understudied (Boix et al., 2012; Miracle et al., 2010).

Limited knowledge and empirical evidence can be overcome by biogeochemical models. Biogeochemical models provide the opportunity to study ecosystems in a more holistic manner (Jørgensen and Bendoricchio, 2011). They can be used as a heuristic tool to identify ecosystem properties and research priorities, articulate hypotheses, and synthesize current knowledge (Duarte et al., 2003; Jørgensen and Bendoricchio, 2011). Biogeochemical models have been widely developed and applied for both deep and shallow lakes (Arhonditsis et al., 2006; Jørgensen, 2010). The development and implementation of biogeochemical models for ponds should ideally capture a set of characteristics that are essential for their biogeochemical functioning. For instance, ecological processes are more intense and complex in ponds than in larger lakes (Downing, 2010). Ponds may lack fish, but could be largely covered by submerged and floating macrophytes, exhibiting a greater terrestrial-aquatic interchange of both organisms and matter (Søndergaard et al., 2005 and references therein). Because of their shallowness and the associated high sediment surface to water volume ratio, an intense water-sediment interaction and fast nutrient turnover can be expected, leading to higher nutrient availability for primary producers than in deeper systems (Meerhoff and Jeppesen, 2009; Søndergaard et al., 2003). Similarly, their small volume makes ponds particularly sensitive to water level fluctuations.

Numerous water quality models have been established to understand and optimize the efficiency of ponds with agricultural or industrial applications; namely, fish production or waste stabilization (Moreno-Grau et al., 1996; Sah et al., 2012; Serpa et al., 2012). A smaller number of biogeochemical models have been employed to artificial ponds to gain insight into eutrophication-related processes. For instance, a model coupling phytoplankton and zooplankton dynamics to water quality was developed for a eutrophic pond created for irrigation in an agricultural area of Japan (Nguyen et al., 2010). Mukherjee et al. (2008) established an ecological model for an artificial pond ecosystem focused on inorganic carbon transfer rates associated with submersed macrophyte photosynthesis, respiration and decomposition. A further example of a eutrophication model for shallow artificial water bodies, specifically for drainage ditches, is the PCDitch model (Janse, 1998). This model aimed to shed light on the linkage between nutrient loadings and shifts to dominant vegetation types in experimental ditches. To our knowledge, no process-based model has been developed with a focus on the biogeochemical functioning of natural, unmanaged pond ecosystems such as kettle holes.

In this study, our main objective is to develop a process-based model that reproduces the biogeochemical processes determining the seasonal dynamics of nutrients, phytoplankton and free-floating macrophytes in a typical kettle hole in Northeast Germany, known as Rittgarten. We first identify the most influential model parameters for the model state variables and quantify the simulated nutrient fluxes associated with the biogeochemical processes considered in our modeling exercise. This process characterization allows us to assess the potential role of kettle holes, with structural and functional similarities to Rittgarten kettle hole, and determine to what extent they act as C sinks or sources to the atmosphere. Finally, we identify knowledge gaps that could guide future research in order to advance our understanding of kettle hole biogeochemistry.

2. Materials and methods

2.1. Study site and data collection

The Rittgarten kettle hole (hereafter Rittgarten) was chosen as case study for this work. Rittgarten (E013°, 42′, 09″, N 53°, 23′, 22″) is located in Uckermark, Northeast Germany, and is a representative system of small shallow shore overflow kettle holes with semi-permanent hydroperiod, which abound in the region (Fig. 1). According to the hydrogeomorphic classification developed by Kalettka and Rudat (2006), these type of kettle holes only dry up every few years after a persistent precipitation deficit. Rittgarten (Fig. 1) has been intensively studied as part of the long-term studies of the LandScales project (Premke et al., 2016; Nitzsche et al. 2017). Its topographic catchment (≈5.7 ha) is dominated by intensive agriculture, and is characterized by a low to medium erosion exposure (Kleeberg et al., 2016a). When it is completely inundated, maximal water depth is about 2.3 m at the deepest point, while the surrounding area is around 1460 m^2. Rittgarten is dominated by edge type vegetation (Luthardt and Dreger, 1996); namely, a reed belt of *Phragmites australis* (Cav.) Trin ex Steudel (Kleeberg et al., 2016a). Rittgarten is hydraulically connected to a shallow aquifer with a thick till layer as lower confining bed. The sediments consist of medium to coarse grained glacial sand alternating with silty-loamy layers of various thickness and

Fig. 1. Rittgarten kettle hole, Uckermark, Northeastern Germany (photo by G. Kazanjian).

lenses of weathered till (Merz and Steidl, 2015). Residence time within the kettle hole basins, assessed via isotope data, is usually in the order of a few years (Nitzsche et al., 2017) which is very long compared to the kinetics of internal processes.

Model forcing functions and calibration were mainly based on a dataset available at ZALF Open Research Data (dataset Kleeberg et al., 2018). Briefly, water samples were collected at 0.3 m depth from July 4th 2013 to July 22th 2014 with biweekly to monthly frequency (except in January 2014). Total phosphorus (TP), phosphate (PO_4), total nitrogen (TN), ammonium (NH_4), nitrate (NO_3), total organic carbon (TOC), dissolved organic carbon (DOC) and chlorophyll-*a* (Chl-*a*) were determined using standard methods as described in Kleeberg et al. (2016a). Total inorganic carbon and dissolved inorganic carbon (TIC and DIC) were determined using a TOC and TN analyzer (multi N/C® 3100, Jena Analytics, Germany) with Non-Dispersive Infrared detector (NDIR) according to DIN EN 1484-H3 (DEV, 2003). Total carbon (TC) was then calculated as the sum of TIC and TOC.

Water temperature, dissolved oxygen (DO) and pH were measured in situ at 0.3 m water depth (WTW CellOx® 325 sensor coupled to a ProfiLine Oxi 3205 and SenTix® 81 pH electrode coupled to a ProfiLine pH 3110 portable meter, respectively). Because water level declined markedly during the study period (Kleeberg et al., 2016a), the distance between the sampling depth (0.3 m) and the sediment decreased accordingly. The water volume dynamics of Rittgarten (Fig. 1, Supplementary Material) were based on water level data measured both manually (with weekly to monthly frequency) but also automatically every 30 min by a CS451 pressure transducer (Campbell Scientific, USA9). This information was combined with site-specific bathymetric and topographic data (Kleeberg et al., 2016b). Water temperature was measured every 30 min by a CS451 pressure transducer (Campbell Scientific, USA9). Daily mean photosynthetic active radiation (PAR) was approximated by a sinusoidal function, based on data recorded every 30 min at the CarboZALF experimental area, located ≈ 6 km from Rittgarten. The free-floating macrophyte and submerged macrophyte biomass determined in July 2013 (77 ± 28 g dry weight m^{-2} and 125 ± 42 g dry weight m^{-2}, respectively; Kleeberg et al., 2016a), was assumed to represent the growing season biomass maxima. Similar to Kazanjian et al. (2018) we calculated the biomass seasonal variation by fitting a polynomial curve and considering that biomass was negligible prior to May and after September.

2.2. Model structure and forcing functions

The model targets the reproduction of seasonal patterns of a number of water quality variables and the primary producers, i.e., phytoplankton and free-floating macrophytes in Rittgarten. Mathematical description of the model is presented in Table 1 and model parameters are defined in Table 2. The Rittgarten model partly builds upon the Lake Washington and Albufera de Valencia eutrophication models (Arhonditsis and Brett 2005a, 2005b; Onandia et al., 2015). Postulating spatial homogeneity in the kettle hole, we developed a single compartment model that depicts four elemental cycles: carbon, nitrogen, phosphorus and oxygen. The model considers the interplay among the following state variables: phytoplankton, free-floating macrophytes, DIC, DOC, POC, NH_4, NO_3, organic N (ON), PO_4, organic P (OP) and DO. The external forcing functions of the model comprise submerged macrophytes biomass, daily mean water temperature, daily mean solar PAR, day length, water level, water pH and TIC (Fig. 1, Supplementary Material). There was no external nutrient input via surface run-off associated to snow melt or heavy rain events (Kazanjian et al., 2018; Thomas Kalettka, personal communication). Further, Rittgarten is surrounded by reed belt and presents a fairly flat topography. Therefore, we assumed that potential pulses of matter input from the catchment did not significantly affect biogeochemical processes in the water body at the time scale of our study and no external nutrient inputs were considered in our modeling exercise.

2.3. Model equations

2.3.1. Phytoplankton

Given the lack of specific information on the composition of the phytoplankton community in Rittgarten, the model considers a single generic phytoplankton group. The phytoplankton equation considers biomass production and losses due to mortality and settling. Phytoplankton production is modeled using a multiplicative function that accounts for the influence of temperature, light, and nutrient availability (Cerco and Cole, 1995). Phytoplankton N and P uptake depends on both intracellular and extracellular concentrations and is constrained by upper and lower internal storage capacity levels. Therefore, N and P dynamics within the phytoplankton cells consider luxury uptake (Hamilton and Schladow, 1997), whereas a Monod-type function was used to depict C uptake solely as a function of extracellular

Table 1
Mathematical equations of the biogeochemical model for the Rittgarten kettle hole.

No.	State Variable	Term	Equation
1	**Phytoplankton biomass**	$\frac{dPHY_x}{dt}$	$= growth_{PHY} \times PHY_x$ $- m_{PHY} \times e^{KTPHY(Tx-Tempref)} \times PHY_x$ $- Vsettling_{PHY} \times PHY_x \times (1/z_x)$, where
	Phytoplankton growth rate	$growth_{PHYx}$	$= growthmax_{PHY} \times fnutrient_{PHYx} \times flight_{PHYx} \times fTemperature_{PHYx}$
	Nutrient limitation	$fnutrient_{PHYx}$	$= min\{\varphi IN_{PHYx}, \varphi PO_{4PHYx}, \varphi DIC_{PHYx}\}$
	Phosphate limitation	φPO_{4PHYx}	$= (Pint_{PHYx} - Pmin_{PHY}) / (Pmax_{PHY} - Pmin_{PHY})$
	Intracellular phosphorus content	$\frac{dP\text{int}_{PHY,x}}{dt}$	$= Pup_{PHYx} \times Pfb_{PHYx} - growth_{PHY} \times Pint_{PHYx}$
	Phosphorus uptake	Pup_{PHYx}	$= Pmaxupt_{PHYx} \times (PO_{4x} / (PO_{4x} + KP_{PHY}))$
	Feedback control	Pfb_{PHYx}	$= (Pmax - Pint_{PHYx}) / (Pmax_{PHY} - Pmin_{PHY})$
	Nitrogen limitation	φIN_{PHYx}	$= (Nint_{PHYx} - Nmin_{PHY}) / (Nmax_{PHY} - Nmin_{PHY})$
	Intracellular nitrogen content	$\frac{dN\text{int}_{PHY,x}}{dt}$	$= Nup_{PHYx} \times Nfb_{PHYx} - growth_{PHY} \times Nint_{PHYx}$
	Nitrogen uptake	Nup_{PHYx}	$= Nmaxupt_{PHYx} \times (IN_x / (IN_x + KN_{PHY}))$
	Feedback control	Nfb_{PHYx}	$= (Nmax_{PHY} - Nint_{PHYx}) / (Nmax_{PHY} - Nmin_{PHY})$
	Carbon limitation	φDIC_x	$= DIC_x / (DIC_x + KC_{PHY})$
	Light limitation	$flight_{PHYx}$	$= 2.718 \times (FD / (K_d \times z)) \times (e^{-a1} - e^{-a0})$, where $a0 = (I/Ik_{PHY})e^{-kd \times ZD}$, $a1 = (I/Ik_{PHY})e^{-kd\ (ZD + z)}$ $FD = photoperiod\ (0 \le FD \le 1)$
	Light attenuation	K_{dx}	$= K_d b + K_d Chla \times (PHY_x / (ChlaC_{PHY}))$ $+ K_d DW \times DUCKWEED_x$
	Temperature limitation	$fTemperature_{PHYx}$	$= e^{(-KTgr_{PHY}\ (T_x - Tref_{PHY})^2)}$
2	**Duckweed biomass**	$\frac{dDW_x}{dt}$	$= growth_{DW} \times DW_x$ $- m_{DW} \times e^{KTDW\ (Tx-Tempref)} \times DW_x$, where
	Duckweed growth rate	$growth_{DWx}$	$= growthmax_{DW} \times fnutrient_{DWx} \times flight_{DWx} \times fTemperature_{DWx} \times fdensity_x$
	Nutrient limitation	$fnutrient_{DWx}$	$= min\{\varphi IN_{DWx}, \varphi PO4_{DWx}\}$
	Phosphate limitation	φPO_{4DWx}	$= (Pint_{DWx} - Pmin_{DW}) / (Pmax_{DW} - Pmin_{DW})$
	Intracellular phosphorus content	$\frac{dP\text{int}_{DW,x}}{dt}$	$= Pup_{DWx} \times Pfb_{DWx} - growth_{DW} \times Pint_{DWx}$
	Phosphorus uptake	Pup_{DWx}	$= Pmaxupt_{DWx} \times (PO_{4x} / (PO_{4x} + KP_{DW}))$
	Feedback control	Pfb_{DWx}	$= (Pmax - Pint_{DWx}) / (Pmax_{DW} - Pmin_{DW})$
	Nitrogen limitation	φIN_{DWx}	$= (Nint_{DWx} - Nmin_{DW}) / (Nmax_{DW} - Nmin_{DW})$
	Intracellular nitrogen content	$\frac{dN\text{int}_{DW,x}}{dt}$	$= Nup_{DWx} \times Nfb_{DWx} - growth_{DW} \times Nint_{DWx}$
	Nitrogen uptake	Nup_{DWx}	$= Nmaxupt_{DWx} \times (IN_x / (IN_x + KN_{DW}))$
	Feedback control	Nfb_{DWx}	$= (Nmax_{DW} - Nint_{DWx}) / (Nmax_{DW} - Nmin_{DW})$
	Light limitation	$flight_{DWx}$	$= (I / I_{sat}) \times (DL/MDL)\ I \le I_{sat,}$ else $= 1 \times (DL/MDL)$
	Temperature limitation	$fTemperature_{DWx}$	$= (Tair_x - Tmin_{DW}) / (Tref_{DW} - Tmin_{DW})$ when $Tair_x \le Tref_{DW}$, else $= (Tmax_{DW} - Tair_x) / (Tmax_{DW} - Tref_{DW})$
	Density limitation	$fdensity_x$	$= K_{DWden} / (K_{DWden} + DW_x)$
3	**Particulate organic carbon concentration**	$\frac{dPOC_x}{dt}$	$= Detritus\ C_x$ $- Settling\ C_x \times OC_x / z_x$ $+ SMD_{POCx}$ $- POChydrol_x$
	Organic carbon	OC_x	$= POC_x + DOC_x$
	Particulate carbon detritus	$Detritus\ C_x$	$= (1 - \alpha_{DOCPHY}) \times m_{PHY} \times e^{KTPHY\ (Tx-Tempref)} \times PHY_x$ $+ \alpha_{POCDW} \times m_{DW} \times e^{KTDW\ (Tx-Tempref)} \times DW_x \times CDW_{DW} \times (1/z_x)$
		$Settling\ C_x$	$= (DetritusC_x / OC_x) \times Vbiosettling + (1 - (DetritusC_x / OC_x)) \times Vsettling$
	Temperature limitation for mineralization	$fTempmin_x$	$= e^{(-KTFmin(T_x - Toptmin)^2)}$
	Submerged macrophyte decomposition	SMD_{POCx}	$= \alpha_{POCSM} \times D_{SM} \times Arrh^{(Tx-Tempref)} \times CDW_{SM} \times SM_x \times (1/z_x)$
	Particulate organic carbon hydrolysis	$POChydrol_x$	$= Khy_{POC} \times fTempmin_x \times POC_x$
4	**Dissolved organic carbon concentration**	$\frac{dDOC_x}{dt}$	$= (\alpha_{DOCPHY} + (1 - (\alpha_{DOCPHY} + \alpha_{POCPHY})) \times (KH_{DOCPHY} / (DO_x + KH_{DOCPHY}))) \times m_{PHY} \times$ $e^{KTPHY\ (Tx-Tempref)} \times PHY_x$ $+ \alpha_{DOCDW} \times m_{DW} \times e^{KTDW\ (Tx-Tempref)} \times DW_x \times CDW_{DW} \times (1/z_x)$ $- KDOCmineral_x \times (DO_x) / (DO_x + KH_{DODOC}) \times DOC_x$ $+ SMD_{DOCx}$ $+ POChydrolysis_x$ $+ DOCENDO_x$
	Dissolved organic carbon mineralization rate	$KDOCmineral_x$	$= fTempmin_x \times KCrefmin$
	Submerged macrophyte decomposition	SMD_{DOCx}	$= \alpha_{DOCSM} \times D_{SM} \times Arrh^{(Tx-Tempref)} \times CDW_{SM} \times SM_x \times (1/z_x)$
5	**Dissolved inorganic carbon concentration**	$\frac{dDIC_x}{dt}$	$= \alpha_{DICPHY} \times m_{PHY} \times e^{KTPHY\ (Tx-Tempref)} \times PHY_x$ $- growthPHY \times PHY_x$ $+ K_{CO2} \times ([CO_2water]_x - [H_2CO_3^*]_x) \times (1/z_x)$ $+ KDOCmineral_x \times (DO_x) / (DO_x + KHdodoc)\ x\ DOC_x$ $+ SMD_{DICx}$ $+ DICENDO_x$
	Submerged macrophyte decomposition	SMD_{DICx}	$= \alpha_{DICSM} \times D_{SM} \times Arrh^{(Tx-Tempref)} \times CDW_{SM} \times SM_x \times (1/z_x)$
	Concentration of dissolved CO_2 at saturation	$[CO_2water]_x$	$= K_{Hx} \times P_{CO2Atmx}$
		K_{Hx}	$= 44000 \times 10^{(0.0158642 \times TKx - 14.0184 + 2385.73 \times TKx)}$, where $TK_x = Temperature\ (K)$
	Concentration of carbonic acid + dissolved CO_2	$[H_2CO_3^*]_x$	$= TIC_x / (1 + (K_1 / [H^+]_x) + ((K_1 + K_2) / [H^+]^2_x)$, where

Table 1 (*continued*)

No.	State Variable	Term	Equation
		K_{1x}	$= 10^{(14.8435\ -\ (14.0184 \times TKx)\ +\ (3404.71 \times TKx))}$
		K_{2x}	$= 10^{(6.498\ -\ (0.02379 \times TKx) + (2902.39 \times TKx))}$
	Endogenous flux	$DICENDO_x$	$= DIC_{sedx} / z_x$
6	**Phosphate concentration**	$\frac{dPO_{4x}}{dt}$	$= -Pup_{PHYx} \times Pfb_{PHYx} \times PHY_x$ $+ \alpha_{PO4PHY} \times m_{PHY} \times e^{KTPHY\ (Tx\text{-}Tempref)} \times Pint_{PHYx} \times PHY_x$ $- Pup_{DWx} \times Pfb_{DWx} \times DW_x \times CDW_{DW} \times (1/z_x)$ $+ KPmineral_x \times OP_x$ $+ SMD_{PO4x}$ $+ PO_4ENDOG_x$
	Phosphorus mineralization rate	$KPmineral_x$	$= fTempmin_x \times KPrefmin$
	Submerged macrophyte decomposition	SMD_{PO4x}	$= \alpha_{PO4SM} \times D_{SM} \times Arrh^{\ (Tx\text{-}Tempref)} \times PDW_{SM} \times SM_x \times (1/z_x)$
	Endogenous flux	PO_4ENDOG_x	$= PO_{4\ sedx} / z_x$
7	**Organic phosphorus concentration**	$\frac{dOP_x}{dt}$	$= DetritusP_x$ $- KPmineral_x \times OP_x$ $- SettlingP \times OP_x / z_x$ $+ SMD_{OPx}$ $+ OPENDO_x$
	Biogenic organic phosphorus accumulation	$DetritusP_x$	$= (1 - \alpha_{PO4PHY}) \times m_{PHY} \times e^{KTPHY\ (T\ -\ Tempref)} \times Pint_{PHYx} \times PHY_x$ $+ \alpha_{OPDW} \times m_{DW} \times e^{KTDW\ (T\ -\ Tempref)} \times Pint_{DWx} \times DW_x \times CDW_{DW} \times (1/z_x)$
	Loss due to particulate phosphorus settling	$SettlingP_x$	$= (DetritusP_x / OP_x) \times Vbiosettling + (1 - (DetritusP_x / OP_x)) \times Vsettling$
	Submerged macrophyte decomposition	SMD_{OPx}	$= \alpha_{OPSM} \times D_{SM} \times Arrh^{\ (Tx\text{-}Tempref)} \times PDW_{SM} \times SM_x \times (1/z_x)$
8	**Ammonium concentration**	$\frac{dNH_{4x}}{dt}$	$= -prefNH_{4PHYx} \times Nup_{PHYx} \times Nfb_{PHYx} \times PHY_x$ $+ \alpha_{NH4PHY} \times m_{PHY} \times e^{KTPHY\ (T\text{-}\ Tempref)} \times Nint_{PHYx} \times PHY_x$ $- prefNH_{4DWx} \times Nup_{DWx} \times Nfb_x \times DW_x \times CDW_{DW} \times (1/z_x)$ $+ KNmineral_x \times ON_x$ $- Nitrification_x$ $+ SMDNH_{4x}$ $+ NH_4ENDO_x$
	Ammonium preference	$prefNH_{4x}$	$= NH_{4x} / (NH_{4x} + NO_{3x})$
	Nitrogen mineralization rate	$KNmineral_x$	$= KN_{refmin} \times fTempmin_x$
	Nitrification rate	$Nitrification_x$	$= Nitmax \times (DO_x / (DO_x + KHdonit)) \times (NH_{4x} / (KHNH_4nit + NH_{4x})) \times fTempnit_x$
	Temperature limitation	$fTempnit_x$	$= e^{(\text{-KTnitr}(\text{T-Toptnitr})^2)}$
	Submerged macrophyte decomposition	SMD_{NH4x}	$= \alpha_{NH4SM} \times D_{SM} \times Arrh^{\ (Tx\text{-}Tempref)} \times NDW_{SM} \times SM_x \times (1/z_x)$
	Endogenous flux	NH_4ENDOG_x	$= NH_{4sedx} / z_x$
9	**Nitrate concentration**	$\frac{dNO_{3x}}{dt}$	$= -(1 - prefNH_{4PHYx}) \times Nup_{PHYx} \times Nfb_{PHYx} \times PHY_x$ $- (1 - prefNH_{4DWx}) \times Nup_{DWx} \times Nfb_{DWx} \times DW_x \times CDW_{DW} \times (1/z_x)$ $+ Nitrification_x$ $- Denitrification_x$ $+ NO_3ENDOG_x$
	Denitrification rate	$Denitrification_x$	$= Denitmax_{sedx} \times (KHoxresp / (DO_x + KHoxresp)) \times (NO_{3x} / (KHNO_3denit + NO_{3x}))$
	Temperature limitation	$fTempdenit_x$	$= e^{(\text{-KTdenitr}(\text{ T -Topdetnitr})^2)}$
	Endogenous flux	NO_3ENDOG_x	$= NO_{3\ sedx} / z_x$
10	**Organic nitrogen concentration**	$\frac{dON_x}{dt}$	$= Detritus\ N_x$ $- KNmineral_x \times ON_x$ $- Settling\ N_x \times ON_x / z_x$ $+ SMD_{ONx}$ $+ ONENDO_x$
	Biogenic organic nitrogen accumulation	$Detritus\ N_x$	$= (1 - \alpha_{NH4PHY}) \times m_{PHY} \times e^{KTPHY\ (T\text{-}Tempref)} \times Nint_x \times PHY_x$ $+ \alpha_{ONDW} \times m_{DW} \times e^{KTDW\ (Tx\text{-}\ Tempref)} \times Nint_{DWx} \times DW_x \times CDW_{DW} \times (1/z_x)$
	Loss due to particulate organic settling	$Settling\ N_x$	$= (Detritus\ N_x / ON_x \times Vbiosettling + (1 - Detritus\ N_x / ON_x) \times Vsettling)$
	Submerged macrophyte decomposition	SMD_{ONx}	$= \alpha_{ONSM} \times D_{SM} \times Arrh^{\ (Tx\text{-}Tempref)} \times NDW_{SM} \times SM_x \times (1/z_x)$
11	**Dissolved oxygen concentration**	$\frac{dDO_x}{dt}$	$= (1.3 - 0.3 \times prefNH_4) \times growthPHY \times Resp_{DO/C} \times PHY_x$ $+ K_{O2} \times (DO_S - DO_x) \times (1/z_x)$ $- (DO_x / (KH_{DOCPHY} + DO_x)) \times m_{PHY} \times e^{KTPHY\ (T\text{-}Tempref)} \times Resp_{DO/C} \times PHY_x$ $- KDOCmineral_x \times Resp_{DO/C}$ $- DOENDO_x$
	Dissolved oxygen saturation	DO_S	$= 14.652 - (0.41022 \times T_x) + (0.007991 \times T_x^2) - (7.7774 \text{ x } 10^{-5} \times T_x^3)$
	Endogenous flux	$DOENDO_x$	$= DO_{sed} / z_x$
12	**Sediment submodel**		
12.1	**Dissolved inorganic carbon sediment release**	$\frac{dDIC_{sed_x}}{dt}$	$= (1 - \beta_C) \times Cdeposition_x$ $- \alpha s_C \times DIC_{sed} \times e^{KTsed(Tsedx\ -Temprefsed)}$
		$Cdeposition_x$	$= Vsettling \times PHY_x + SettlingC_x \times POC_x$
12.2	**Dissolved oxygen sediment consumption**	$\frac{dDO_{sed_x}}{dt}$	$= (1 - \beta_C) \times Cdeposition_x$ $+ (\alpha sO_2 \times DOsed_x \times e^{KTsed(Tsedx\ -Temprefsed)}$
12.3	**Phosphate sediment release**	$\frac{dPO_{4sed_x}}{dt}$	$= (1 - \beta_P) \times Pdeposition_x$ $- \alpha s_{PO4} \times PO_{4sedx} \times e^{KTsed(Tsed\ -Temprefsed)}$
		$Pdeposition$	$= (Vsettling \times Pint_x \times PHY_x + SettlingP_x \times OP_x)$
12.4	**Ammonium sediment release**	$\frac{dNH_{4sed_x}}{dt}$	$= (1 - \beta_N) \times Ndeposition_x$ $- (\alpha s_{NH4} \times NH_{4sedx} \times e^{KTsed(Tsedx\ -Temprefsed)}))$ $- Nitmax_{sed} \times fTempnitsed_x \times (DO_x / (DO_x + KHdonit_{sed})) \times$ $(NH_{4sedx} / (KHNH_4nit_{sed} + NH_{4\ sedx}))$

(continued on next page)

Table 1 (*continued*)

No.	State Variable	Term	Equation
		$Ndeposition_x$	$=Vsettling_x \times Nint_x \times PHY_x + SettlingN_x \times ON_x$
	Temperature limitation for nitrification in the sediments	$fTempnitsed_x$	$= e^{(-KTnitr_{sed}(T_x - Toptnitr_{sed})^2)}$
12.5	**Nitrate sediment release**	$\frac{dNO_{3sed_x}}{dt}$	$= Nitmax_{sed} \times fTempnit_x \times (DO_x / (DO_x + KHdonit_{sed})) \times (NH_{4sedx} / (KHNH_4nit_{sed} + NH_{4sedx})) - (as_{NO3} \times NO_{3sedx} \times e^{KTsed(Tsedx - Temprefsed)})) - Denitmax_{sed} \times fTempdenitsed_x \times (KHdodenit_{sed} / (DO_x + KHdodenit_{sed})) \times (NO_{3sedx} / KHNO_3denit_{sed} + NO_{3\ sedx})$
	Temperature limitation for denitrification in the sediments	$fTempdenitsed_x$	$= e^{(-KTdenitsed\ (Tx - Toptdenitsed)2)}$
12.6	**Organic nitrogen sediment release**	$ONENDO_x$	$= ONsed_x \times e^{KTsed(Tsed - Temprefsed)}$, where
		$ONsed_x$	$= OPsed \times TN_x/TP_x$
12.7	**Organic phosphorus sediment release**	$OPENDO_x$	$= OPsed \times e^{KTsed(Tsedx - Temprefsed)}$, where
		$OPsed$	$= 0.1\ mg\ m^{-2}\ day^{-1}$
12.8	**Dissolved organic carbon sediment release**	$DOCENDO_x$	$= DOCsed \times e^{KTsed(Tsed - Temprefsed)}$, where
		$DOCsed$	$= 0.03\ mg\ m^{-2}\ day^{-1}$

concentrations. The dependence of phytoplankton growth on temperature is modeled by a function similar to a Gaussian probability curve, with an increase until an optimal temperature above which growth declines (Cerco and Cole, 1995). The relationship between photosynthesis and light intensity is represented by the Steele's equation combined with Beer's law to scale photosynthetically active radiation (PAR) to depth (Arhonditsis and Brett, 2005a). Similar to Mulderij et al. (2007), the extinction coefficient (K_d) is calculated as the sum of the background light attenuation (accounting for water absorption and the attenuation due to total suspended solids), the attenuation due to Chl-*a*, and that due to free-floating macrophytes. Phytoplankton mortality accounts for mortality due to zooplankton grazing and natural causes, as well as all internal processes that reduce algal biomass (respiration, excretion). Phytoplankton mortality is assumed to increase exponentially with temperature in our model. Phytoplankton settling represents the loss of biomass caused by algal sinking to the kettle hole sediment.

2.3.2. Duckweed – free floating macrophytes

Duckweed (DW), i.e. species of Lemnaceae, generally grow forming floating mats over eutrophic waters that are either stagnant or present low velocity (Portielje and Roijackers, 1995). Free-floating macrophytes mats frequently develop from May to September in Rittgarten covering to a variable extent the water surface (Kazanjian et al., 2018). In July 2013, the kettle hole was covered by an indistinguishable mix of the common DW *Lemna minor* L. and *Spirodela polyrhiza* (L.) Schleid. Therefore, free-floating macrophytes were modeled as a single group.

The DW equation accounts for biomass production and losses due to basal metabolism. Biomass production is mathematically expressed by a multiplicative formulation that considers the effects of temperature, light, nutrient availability and crowding on DW growth (Driever et al., 2005; Landolt and Kandeler, 1987). Duckweed are fast growing organisms that have competitive advantage for light and high temperature optima (Landolt and Kandeler, 1987). Consequently, DW is assigned high maximum growth rates and low mortality, fast P and slow N kinetics, and high temperature optima.

Duckweed growth is dependent on both external and internal N and P concentrations (Kufel et al., 2012; Portielje and Roijackers, 1995). Consequently, DW N and P uptake is defined as a function of both ambient and internal nutrient concentrations, constrained by upper and lower tissue storage capacity limits (Janse and Van Puijenbroek, 1997). Carbon is not considered a limiting nutrient, since DW fronds are in direct contact with the atmosphere. The influence of temperature on DW growth rate is formulated as a linear relationship from the minimum temperature required for growth until an optimal temperature. When water temperature surpasses the optimal temperature for DW growth, our model postulates a linear decrease (Peeters et al., 2013 and references therein). Photosynthesis of DW is assumed to increase linearly in response to light intensity up to a saturation level and in direct proportion to day length (Jassby and Platt, 1976; Peeters et al., 2013). The crowding (plant mat density) limitation is depicted by a Monod-like function (Driever et al., 2005). Basal metabolism of DW considers mortality due to natural causes besides those internal processes that reduce algal biomass (respiration, excretion) and is assumed to increase exponentially with temperature.

2.3.3. Carbon

The model considers three state variables: DIC, DOC and POC. The DIC equation accounts for phytoplankton uptake, the fraction of phytoplankton mortality that is released into the water column, the bacteria-mediated mineralization of DOC and the fraction of submerged macrophyte decomposition released as DIC. The submerged macrophyte decomposition term considers the fraction of nutrients that returns into the water column through decomposition and is assumed to be proportional to the observed biomass, while accounting for temperature dependence by using the Arrhenius equation. Finally, the DIC equations considers atmospheric exchange. The latter process is modeled as a function of the CO_2 transfer velocity and the gradient between the realized surface water CO_2 concentration and the corresponding CO_2 concentration at saturation (Xiao et al., 2014). The CO_2 saturation concentration was calculated based on Henry's law constant (Bowie et al., 1985). The DOC equation includes the fraction of phytoplankton and duckweed mortality released into the water column, the fraction of POC that is hydrolyzed to DOC, the losses due to bacteria-mediated mineralization and the fraction of submerged macrophyte decomposition released as DOC. The POC equation considers the contributions from phytoplankton and duckweed basal metabolism as well as from submerged macrophyte decomposition. Part of the POC is hydrolyzed to DOC and another fraction settles to the sediment.

2.3.4. Nitrogen

The model considers three N forms: NH_4, NO_3 and ON. The NH_4 equation considers uptake by phytoplankton and DW, the fraction of phytoplankton mortality that is excreted into the water column, the bacteria-mediated mineralization of ON and the fraction of submerged macrophyte decomposition released as NH_4. The NO_3 equation accounts for uptake by phytoplankton and DW as well as the fraction of submerged macrophyte decomposition released as NO_3. The effect of inhibition of NO_3 uptake by NH_4 is derived from the relative abundance of the two inorganic N forms. Under oxygenated conditions, NH_4 is oxidized to NO_3 through nitrification. This process is modeled as a function of NH_4, DO and temperature in the water column (Cerco and Cole, 1995). Under oxygen-depleted conditions, NO_3 is lost from the system via denitrification. Denitrification is formulated as a function of NO_3, DO and temperature availability (Arhonditsis and Brett, 2005a). The ON equation considers the contributions from phytoplankton and DW basal metabolism as well as from submerged macrophyte

Table 2
Definitions and calibration values of the model parameters.

Symbol	Description	Values	Units	Sources
Arrh	Arrhenius constant	1.047	–	*
$ChlaC_{PHY}$	Chlorophyll-*a* to carbon ratio for phytoplankton	0.02	–	8, 9, 11, 14
CDW_{DW}	Carbon to dry weight ratio for duckweed	0.38	mg C mg d w^{-1}	1
CDW_{SM}	Carbon to dry weight ratio for submerged macrophyte	0.38	mg C mg d w^{-1}	1
D_{SM}	Submerged macrophyte decomposition rate	0.009	d^{-1}	*
$Denitmax_{sed}$	Maximum denitrification rate in the sediment	20	mg N m^{-2} d^{-1}	8
$growthmax_{DW}$	Maximum growth rate for duckweed	0.38	day^{-1}	2, 4, 5
$growthmax_{PHY}$	Maximum growth rate for phytoplankton	1.33	day^{-1}	13, 14
I_{sat}	Saturation light intensity for duckweed	460	μmol m^{-2} s^{-1}	6
Ik_{PHY}	Half saturation light intensity for phytoplankton	110	μmol m^{-2} s^{-1}	*
KCrefmineral	Carbon mineralization rate at reference temperature	0.015	day^{-1}	10
K_db	Background light attenuation	0.01	m^{-1}	14
K_dChla	Light attenuation coefficient for chlorophyll-*a*	0.015	m^2 mg^{-1}	14
K_dDW	Light attenuation coefficient for duckweed	0.0002	m^2 mg^{-1}	*
K_{DWden}	Limiting coefficient for duckweed density	45	g DW m^{-2}	6
K_{CO2}	CO_2 atmospheric exchange rate	0.0585	m d^{-1}	7
KC_{PHY}	Half saturation constant for carbon uptake by phytoplankton	2000	mg C m^{-3}	13
KH_{DOCPHY}	Half saturation concentration for DOC excretion by phytoplankton	8	mg O_2 m^{-3}	*
KHdoDOC	Half saturation concentration of DO required for oxic mineralization	0.5	mg O_2 m^{-3}	10
$KHdodenit_{sed}$	Half saturation concentration of DO deficit required for denitrification in the sediments	0.5	mg O_2 m^{-3}	*
KHdonit	Half saturation concentration of DO required for nitrification	1	mg O_2 m^{-3}	10
$KHdonit_{sed}$	Half saturation concentration of DO required for nitrification in the sediments	0.5	mg O_2 m^{-3}	*
$KHNH_4nit$	Half saturation concentration of NH_4 required for nitrification	0.08	mg N m^{-3}	10
$KHNH_4nit_{sed}$	Half saturation concentration of NH_4 required for nitrification in the sediments	0.5	mg N m^{-3}	*
$KHNO_3denit$	Half saturation concentration of NO_3 required for denitrification	0.15	mg N m^{-3}	*
$KHNO_3denit_{sed}$	Half saturation concentration of NO_3 required for denitrification in the sediments	0.03	mg O_2 m^{-3}	*
Khy_{POC}	POC hydrolysis coefficient	0.3	–	21*
KN_{DW}	Half saturation constant for nitrogen uptake by duckweed	100	mg N m^{-3}	17*, 23*, 24*
KN_{PHY}	Half saturation constant for nitrogen uptake by phytoplankton	75	mg N m^{-3}	14
KNrefmineral	Nitrogen mineralization rate at reference temperature	0.04	day^{-1}	14
K_{O2}	Reaeration coefficient	0.065	m day^{-1}	*
KP_{PHY}	Half saturation constant for phosphorus uptake by phytoplankton	25	mg P m^{-3}	14
KP_{DW}	Half saturation constant for phosphorus uptake by duckweed	20	mg P m^{-3}	17*, 23*, 24*
KPrefmineral	Phosphorus mineralization rate at reference temperature	0.08	day^{-1}	14
KTdenit	Effect of temperature on denitrification	0.004	ºC^{-2}	*
KT_{DW}	Effects of temperature on duckweed mortality	0.001	ºC^{-1}	*
KT_{PHY}	Effects of temperature on phytoplankton mortality	0.002	ºC^{-1}	*
KTFmin	Effects of temperature on mineralization	0.0025	ºC^{-2}	*
$KTdenit_{sed}$	Effect of temperature on sediment denitrification	0.004	ºC^{-2}	*
KTnit	Effect of temperature on nitrification	0.004	ºC^{-2}	10,15
$KTnit_{sed}$	Effect of temperature on sediment nitrification	0.004	ºC^{-2}	*
$KTgr_{PHY}$	Effect of temperature on phytoplankton	0.005	ºC^{-2}	3, 10, 12, 13
KT_{sed}	Effect of temperature on sediment-water flux	0.004	ºC^{-1}	*
m_{DW}	Mortality rate for duckweed	0.031	day^{-1}	5
m_{PHY}	Mortality rate for phytoplankton	0.016	day^{-1}	5
NDW_{SM}	Nitrogen to dry weight ratio in submerged macrophyte	0.025	mg N mg d w^{-1}	1
Nitmax	Maximum nitrification rate at optimal temperature	0.5	mg N m^{-3} day^{-1}	22
$Nitmax_{sed}$	Maximum sediment nitrification rate	0.5	mg N m^{-2} day^{-1}	10, 15, 19
$Nmax_{DW}$	Maximum duckweed internal nitrogen	0.05	mg N mg d w^{-1}	5*
$Nmax_{PHY}$	Maximum phytoplankton internal nitrogen	0.04	mg N mg C^{-1}	14
$Nmaxupt_{DW}$	Maximum nitrogen uptake rate for duckweed	0.011	mg N mg dw^{-1} day^{-1}	5
$Nmaxupt_{PHY}$	Maximum nitrogen uptake rate for phytoplankton	0.02	mg N mg C^{-1} day^{-1}	14*
$Nmin_{PHY}$	Minimum phytoplankton internal nitrogen	0 0.02	mg N mg C^{-1}	14
$Nmin_{DW}$	Minimum duckweed internal nitrogen	0.02	mg N mg d w^{-1}	5
P_{CO2Atm}	CO_2 partial atmospheric pressure	3.6×10^{-4}	atm	*
PDW_{SM}	Phosphorus to dry weight ratio for submerged macrophyte	0.007	mg P mg d w^{-1}	1
$Pmax_{DW}$	Maximum duckweed internal phosphate	0.001	mg P mg d w^{-1}	5*
$Pmax_{PHY}$	Maximum phytoplankton internal phosphate	0.25	mg P mg C^{-1}	14
$Pmaxupt_{DW}$	Maximum phosphorus uptake rate for duckweed	0.002	mg P mg d w^{-1} day^{-1}	5
$Pmaxupt_{PHY}$	Maximum phosphorus uptake rate for phytoplankton	0.012	mg P mg C^{-1} day^{-1}	14
$Pmin_{DW}$	Minimum duckweed internal phosphorus	0 0.0009	M g P mg d w^{-1}	5*
$Pmin_{PHY}$	Minimum phytoplankton internal phosphorus	0.02	M g P mg C^{-1}	14
$Resp_{DO/C}$	Dissolved oxygen to carbon ratio in respiration	2.67	mg O_2 mg C^{-1}	10, 14, 16
Tempref	Water reference temperature	20	ºC	3, 7, 10, 11
$Tempref_{sed}$	Sediment reference temperature	20	ºC	*
$Tmin_{DW}$	Minimum temperature required for duckweed growth	5	ºC	6, 17
$Tmax_{DW}$	Maximum temperature for duckweed growth	38	ºC	17, 26
Toptdenit	Optimal temperature for denitrification	20	ºC	*
$Toptdenit_{sed}$	Optimal temperature for denitrification in sediment	20	ºC	*
Toptmin	Optimal temperature for mineralization	20	ºC	*
Toptnit	Optimal temperature for nitrification	20	ºC	10,15
$Toptnit_{sed}$	Optimal temperature for nitrification in sediment	20	ºC	*
$Tref_{DW}$	Reference temperature for duckweed metabolism	26	ºC	2, 6, 17, 18
$Tref_{PHY}$	Reference temperature for phytoplankton metabolism	18	ºC	20

(continued on next page)

Table 2 (*continued*)

Symbol	Description	Values	Units	Sources
Vbiosettling	Biogenic particle settling velocity	1.5	m day^{-1}	1
Vsettling	Allochthonous particle settling velocity	1.5	m day^{-1}	1
$Vsettling_{PHY}$	Phytoplankton settling velocity	0.0165	m day^{-1}	10, 16
ZD	Distance from the water surface to the top of model segment	0	m	*
α_{DICPHY}	Fraction of phytoplankton mortality becoming dissolved inorganic carbon	0.15	–	10
α_{DICSM}	Fraction of submerged macrophyte decomposition released as dissolved inorganic carbon	0.01	–	*
α_{DOCDW}	Fraction of duckweed mortality becoming dissolved organic carbon	0.1	–	*
α_{DOCPHY}	Fraction of phytoplankton mortality becoming dissolved organic carbon	0.1	–	10
α_{DOCSM}	Fraction of submerged macrophyte decomposition released as dissolved organic carbon	0.29	–	*
α_{NH4PHY}	Fraction of phytoplankton mortality becoming ammonium	0.08	–	10
α_{NH4SM}	Fraction of submerged macrophyte decomposition released as ammonium	0.3	–	*
α_{ONDW}	Fraction of duckweed mortality becoming organic nitrogen	0.65	–	*
α_{ONSM}	Fraction of submerged macrophyte decomposition released as organic nitrogen	0.7	–	*
α_{OPDW}	Fraction of duckweed mortality becoming organic phosphorus	0.8	–	*
α_{OPSM}	Fraction of submerged macrophyte decomposition released as organic phosphorus	0.5	–	*
α_{PO4PHY}	Fraction of phytoplankton mortality becoming phosphate	0.85	–	10
α_{PO4SM}	Fraction of submerged macrophyte decomposition released as phosphate	0.5	–	*
α_{POCSM}	Fraction of submerged macrophyte decomposition released as particulate organic carbon	0.7	–	*
αs_C	Sediment carbon release rate	0.01	day^{-1}	*
αs_{NO3}	Sediment nitrate release rate	0.03	day^{-1}	*
αs_{NH4}	Sediment ammonium release rate	0.06	day^{-1}	*
αs_{PO4}	Sediment phosphate release rate	0.002	day^{-1}	*
αsO_2	Sediment dissolved oxygen consumption	15	day^{-1}	*
βc	Fraction of inert carbon buried into deeper sediment	0.5	–	*
β_N	Fraction of inert nitrogen buried into deeper sediment	0.65	–	*
β_P	Fraction of inert phosphorus buried into deeper sediment	0.75	–	*

1) Kleeberg et al. (2016a); 2) Driever et al. (2005); 3) Omlin et al. (2001); 4) Lüönd (1983); 5) Janse and Van Puijenbroek (1997); 6) Peeters et al. (2013) and references therein; 7) Xiao et al. (2014); 8) Wetzel (2001); 9) Chen et al. (2002) and references therein; 10) Cerco and Cole (1994) and references therein; 11) Mian and Yanful (2004); 11) Reynolds (1984); 12) Arhonditsis and Brett (2005); 13) Reynolds (2006); 14) Hamilton and Schladow (1997) and references therein; 15) Berounsky and Nixon (1990); 16) Bowie et al. (1985); 17) Lasfar et al. (2007); 18) Van der Heide et al. (2006); 19) Lampert and Sommer (2007); 20) Chen (2015); 21) Nguyen et al. (2010); 22) Small et al. (2013); 23) Schmitt et al. (2013); 24) Lüönd (1980);
* Subject to calibration.

decomposition. Part of ON settles to the sediment and another fraction is mineralized by bacteria to NH_4.

2.3.5. Phosphorus

The model considers two P state variables: PO_4 and OP. The PO_4 equation includes phytoplankton and floating macrophytes uptake, the fraction of phytoplankton mortality that is directly released into the system in inorganic form, the bacteria-mediated mineralization of OP and the fraction of submerged macrophyte decomposition released as PO_4. The OP equation accounts for the fraction of OP that is returned into the water column via phytoplankton and DW mortality as well as from submerged macrophyte decomposition. Part of the OP settles to the sediment and another fraction is mineralized by bacteria to PO_4.

2.3.6. Dissolved oxygen

The DO equation includes phytoplankton photosynthesis and atmospheric reaeration as major sources, and phytoplankton respiration, organic matter (i.e., DOC) heterotrophic respiration as well as nitrification as major sinks. In the model, reaeration is proportional to the difference between DO saturation (DO_s) and the actual DO concentration (Bowie et al., 1985). Dissolved oxygen saturation is assumed to be a function of temperature and was calculated based on the polynomial equation proposed by Elmore and Hayes (1960).

2.3.7. Fluxes from the sediment

The available empirical data did not enable the addition of a detailed sediment diagenesis submodel. Consequently, the interactions between the water column and the sediment are represented by relating temperature-dependent N and P fluxes as well as sediment oxygen consumption to sedimentation and burial rates (Arhonditsis and Brett, 2005a). Additionally, nitrification occurring at the sediment surface defines the relative proportion of NH_4 and NO_3 fluxes. Because Rittgarten is protected from wind by a reed belt, we assumed that wind-driven resuspension of the nutrient organic fractions settled onto the sediment was negligible. In agreement with this assumption, Kleeberg et al. (2016a) ruled out significant resuspension events in Rittgarten during the time period considered by our modeling exercise.

2.4. Model analysis

The model was calibrated to match the observed data from July 2013 to July 2014. To achieve this goal, the model parameters were manually tuned within the parameter value ranges reported in the literature. The calibration vector which provided the best fit between simulated and observed values of the state variables is presented in Table 2. The goodness-of-fit between the simulated and observed values was assessed based on four criteria: Kling-Gupta efficiency (KGE), root mean squared error (RMSE), relative error (RE) and average error (AE). The KGE evaluates the overall match between observations and model predictions based on an equal weighting of correlation, bias, and variability, with an ideal value of unity (Gupta et al., 2009). The RMSE, RE and AE assess the accuracy of model predictions and should be ideally zero. Whereas the RMSE is an absolute measure of model fit, the RE is a scale-independent goodness-of-fit statistic of the percentage discrepancy between model predictions and observations, and the AE reflects the model bias towards under or overestimating the observations (Bennett et al., 2013).

In order to evaluate the effect of model parameters changes on the model results, a relative sensitivity measure (RS_i) was calculated. Every model parameter was increased and decreased by 10% and the influence on phytoplankton, DW, DOC, POC, DIC, TC, NH_4, NO_3, TN, PO_4, TP and DO, was calculated as described in Zhang and Rao (2012):

$$RS_i = (\Delta Y_i / Y_i)/(\Delta\theta/\theta)$$

where Y_i is the daily value of the state variable, ΔY_i represents the change in the modeled value, θ is the calibrated parameter value and $\Delta\theta$ indicates the modification applied to the parameter. The mean annual relative sensitivity ($MARS_i$) was then calculated.

3. Results

3.1. Model predictions versus observed dynamics

Observations and model predictions during July 2013 to July 2014 are in reasonably good accordance (Fig. 2). The model adequately predicts the observed phytoplankton seasonal cycle. The Chl-*a* minimum ($\approx$3 μg L^{-1}) and maximum ($\approx$80 μg L^{-1}) concentrations in our calibration dataset are reasonably captured, albeit the early summer peak in 2014 is substantially underestimated. The expected seasonal DW dynamics is realistically reproduced, including DW absence during winter, but underestimates the observed DW maximum towards the end of June/mid-July. The model successfully predicts the annual trend of DOC but is not capable of fully capturing the increase in spring and subsequent peak in summer. The POC predictions lie within the observed concentration range. However, the observed seasonal dynamics are not successfully simulated by the model, with a marked underprediction of the observed maxima at the beginning of the spring ($\approx$1 mg L^{-1} vs $\approx$6 mg L^{-1}). The ranges of DIC and TC predictions (calculated as the sum of DOC, POC and DIC) accurately correspond to those of the observations. Their seasonal dynamics are successfully reproduced by the model, apart from a marked pulse recorded in winter. The model is not capable of reproducing the observed NH_4 and NO_3 trends and overestimates the observed concentrations. It should be noted though that most NO_3 observed concentrations are close to the detection limit (0.03 mg L^{-1}) and represent a minor fraction of TN. The TN predictions underestimate the measured concentrations, especially during summer and autumn. The model reasonably reproduces the seasonal variability of PO_4 and TP, but underestimates the concentrations in summer ($\approx$0.3–0.6 mg L^{-1} vs $\approx$0.2 mg L^{-1}). The observed seasonal dynamics of DO are reasonably reproduced by the model. Nonetheless, the model overpredicts the measured concentrations during the winter period ($\approx$9 mg L^{-1} vs $\approx$6 mg L^{-1}) and late spring. During the calibration period, the maximum Kling-Gupta efficiency was obtained for PO_4. Values above 0.18 were found for phytoplankton, DIC, TP and DO. However, DOC, POC, TC, NH_4, NO_3, and TN had a negative modeling efficiency (Table 3).

3.2. Nitrogen, phosphorus, oxygen and carbon budget

The simulated annual C, N and P cycles as well as the magnitude of their cumulative annual fluxes associated with the biogeochemical processes considered by the model are shown in Fig. 3. Floating macrophytes and phytoplankton net growth (uptake minus basal metabolism) collectively consume 0.41 kg y^{-1} of P and 2.17 kg y^{-1} of N. Phytoplankton utilize 0.35 kg y^{-1} of P for net growth, which represents $\approx$85% of the P uptake for net growth by primary producers, and 0.21 kg y^{-1} of N; that is, 10% of the consumed N. Predicted bacteria-

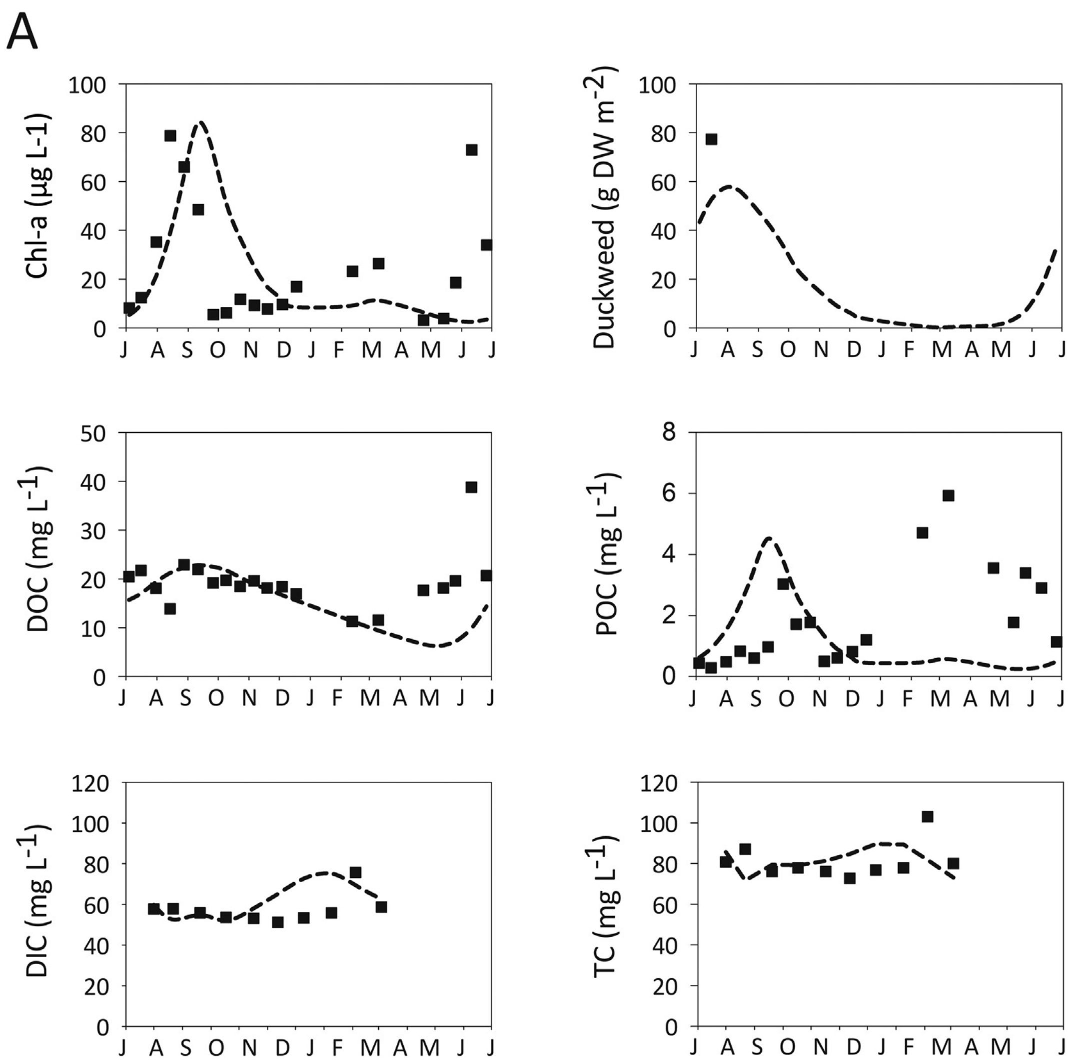

Fig. 2. Comparison between observed and simulated values for (A) phytoplankton, duckweed, dissolved organic carbon (DOC), particulate organic carbon (POC), dissolved inorganic carbon (DIC), total carbon (TC); (B) ammonium (NH_4), nitrate (NO_3), total nitrogen (TN), phosphate (PO_4), total phosphorus (TP) and dissolved oxygen (DO) during the period July 2013 to July 2014 in the Rittgarten kettle hole. Black squares represent the observed data and dashed lines represent the model predictions.

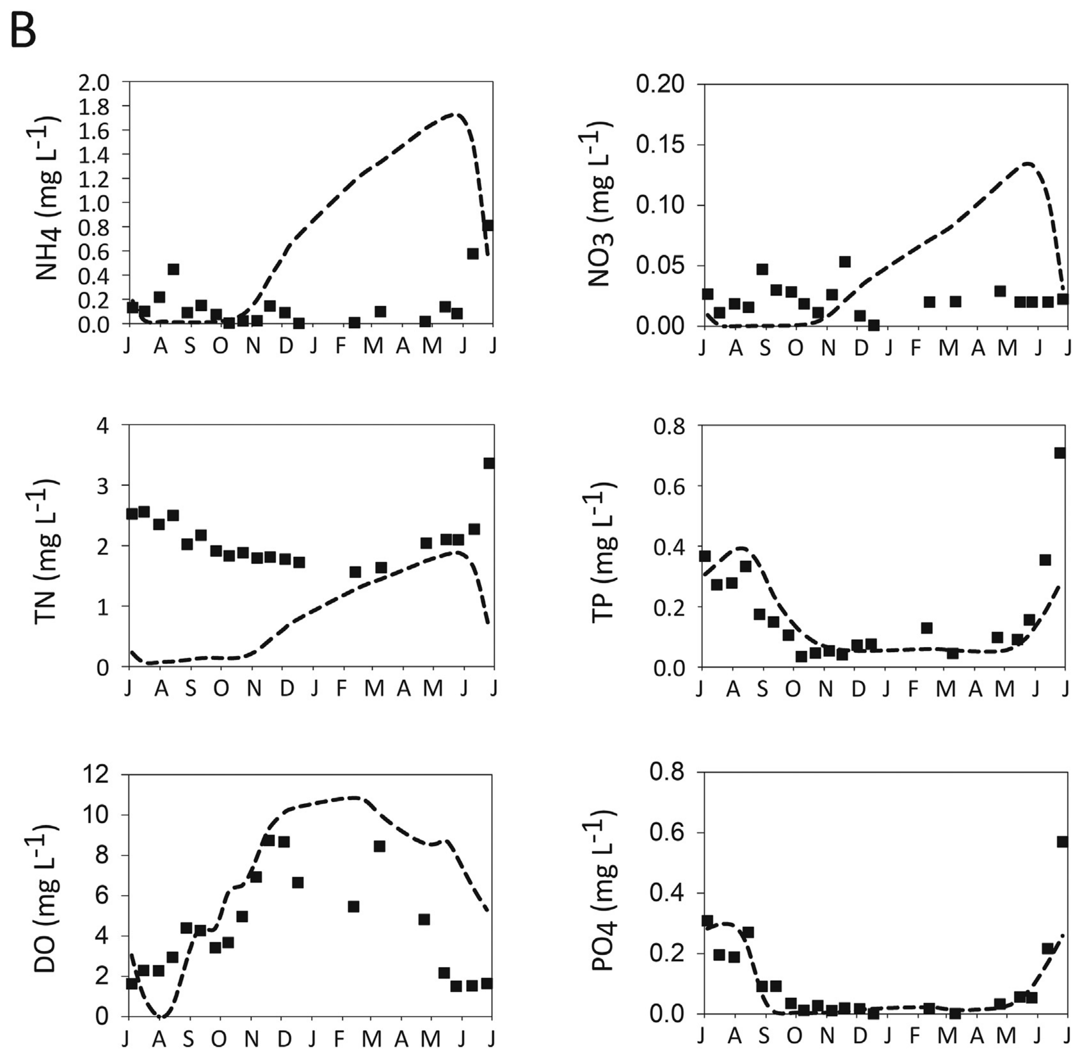

Fig. 2 (*continued*).

mediated mineralization combined with phytoplankton basal metabolism add 0.13 kg y^{-1} of P, which is not sufficient to meet the floating macrophytes and phytoplankton P requirements (0.53 kg y^{-1}). Similarly, the joint N demand of floating macrophytes and phytoplankton (2.18 kg y^{-1}) is not met by these processes (0.05 kg y^{-1}). Submerged macrophyte decomposition contributes 0.19 kg y^{-1} and 0.4 kg y^{-1} of the P and N, which represents 36% and 18% of the P and N requirements, respectively. In quantitative terms, the major process affecting the DIC concentrations is CO_2 exchange with the atmosphere (45.35 kg y^{-1}). With regard to the sediment flux dynamics, the model is suggestive of an equilibrium between processes that act as N and P sources and sinks at the annual scale. According to our N model predictions, 5.45 kg y^{-1} are deposited on the sediments and 5.14 kg y^{-1} get back into the water column, mainly in organic form. In the case of P, 0.39 kg y^{-1} is settled on the sediments and 0.32 kg y^{-1} is released into the water column, predominantly in inorganic form. Finally, the model predicts a C deposition of 55.56 kg y^{-1} from which 33.3 kg y^{-1} return to the water column, pointing to burial of 40% of the C settled on the sediment.

Table 3
Goodness-of-fit statistics for Rittgarten kettle hole model state variables for the calibration period: Kling-Gupta efficiency (KGE), root mean squared error (RMSE), relative error (RE) and average error (AE).

State variables	K-G	RMSE	RE	AE
Chlorophyll-*a* (μg L^{-1})	0.25	29.15	0.84	−0.73
DO (mg L^{-1})	0.30	3.27	0.62	1.90
POC (mg L^{-1})	−0.29	2.29	0.95	−0.44
DIC (mg L^{-1})	0.19	10.31	0.14	4.94
DOC (mg L^{-1})	−0.02	8.42	0.27	−3.49
TC (mg L^{-1})	−0.39	13.15	0.14	−2.48
NH_4 (mg L^{-1})	−2.54	0.79	3.45	0.43
NO_3 (mg L^{-1})	−2.34	0.05	1.84	0.02
PO_4 (mg L^{-1})	0.67	0.08	0.42	−0.02
TN (mg L^{-1})	−0.50	1.64	0.67	−1.41
TP (mg L^{-1})	0.59	0.12	0.44	−0.01

3.3. Sensitivity analysis

The sensitivity analysis assessed the influence of all model parameters on the model state variables. The ten parameters that had the highest positive and negative impact on the state variables are presented in Table 4. These parameters are related to the DO to C ratio in respiration ($Resp_{DO/C}$), submerged macrophyte decomposition (D_{SM}), the fraction of N and P buried into deep sediment (β_N, β_P) and growth and losses of primary producers. In particular, the parameters related to growth and losses of primary producers are: maximum growth rate for DW, DW reference temperature, minimum temperature required for DW growth and DW mortality rate, maximum growth rate for phytoplankton, half saturation light intensity for phytoplankton photosynthesis, reference temperature for phytoplankton metabolism and phytoplankton settling velocity ($growthmax_{DW}$, $Tref_{DW}$, $Tmin_{DW}$, m_{DW}, $growthmax_{PHY}$, Ik_{PHY}, $Tref_{PHY}$, $Vsettling_{PHY}$).

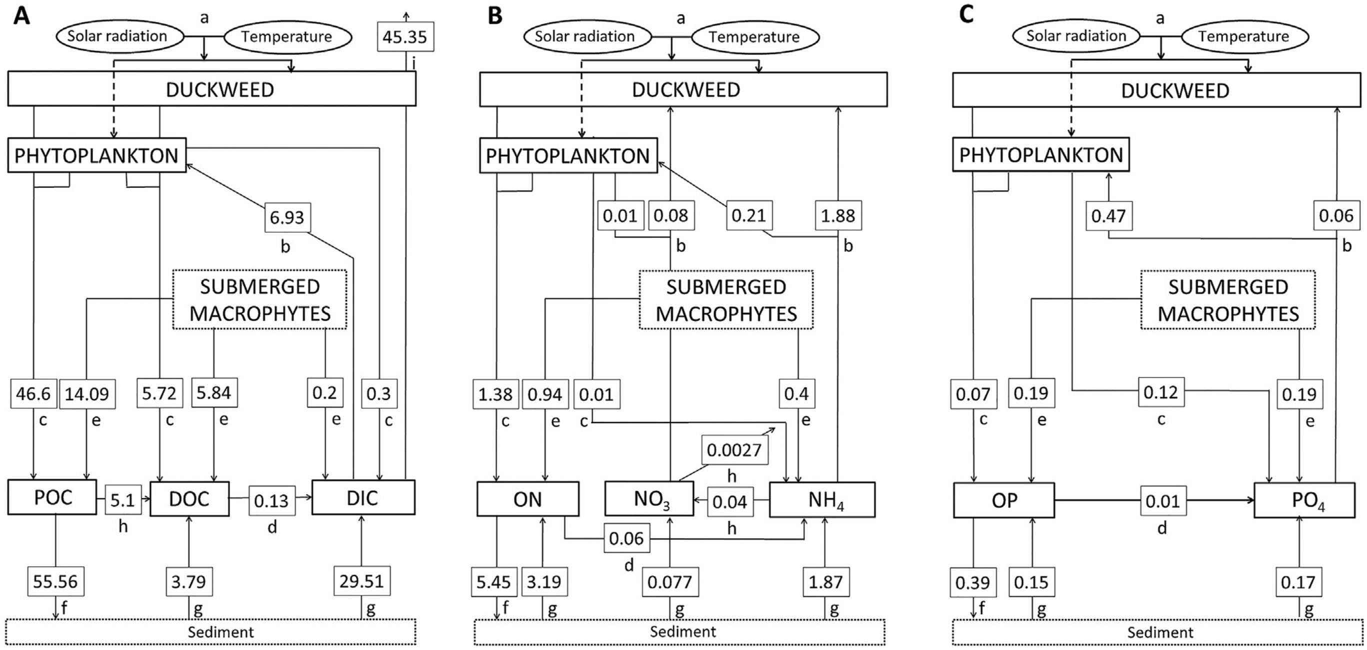

Fig. 3. Biogeochemical processes simulated by the Rittgarten kettle hole model. (A) Carbon cycle: (a) external forcing to duckweed and phytoplankton growth (temperature, solarradiation); (b) uptake by phytoplankton; (c) phytoplankton basal metabolism excreted as dissolved inorganic carbon (DIC), dissolved organic carbon (DOC) and particulate organic carbon (POC); (d) POC mineralization; (e) fraction of submerged macrophyte decomposition released as DIC, DOC and POC; (f) particles settling; (g) DIC and DOC fluxes from sediment driven by diffusion; (h) POC hydrolysis; (i) DIC losses due to atmospheric exchange; (B) Nitrogen cycle: (a) external forcing to duckweed and phytoplankton growth (temperature, solar radiation); (b) uptake by duckweed and phytoplankton; (c) duckweed and phytoplankton basal metabolism excreted as ammonium (NH_4) or/and organic nitrogen (ON); (d) ON mineralization; (e) fraction of submerged macrophyte decomposition released as NH_4 and ON; (f) particles settling; (g) NH_4, nitrate (NO_3) and ON fluxes from sediment driven by diffusion; (h) NO_3 losses due to de-/nitrification; (C) Phosphorus cycle: (a) external forcing to duckweed and phytoplankton growth (temperature, solar radiation); (b) uptake by duckweed and phytoplankton; (c) duckweed and phytoplankton basal metabolism excreted as phosphate (PO_4) and organic phosphorus (OP); (d) OP mineralization; (e) fraction of submerged macrophyte decomposition released as PO_4 and OP; (f) particles settling; (g) PO_4 and OP fluxes from sediment driven by diffusion. Given numbers represent cumulative annual nitrogen, phosphorus and carbon fluxes (kg y^{-1}) for the period July 2013 to July 2014.

The parameters having the strongest impact on phytoplankton are those affecting the DO to C ratio in respiration ($Resp_{DO/C}$), P burial and uptake (β_P, $Pmax_{PHY}$), as well as growth and losses ($Vsettling_{PHYT}$, $growthmax_{PHYT}$, Ik_{PHYT} and $Tref_{PHYT}$). Biomass of DW is mostly influenced by parameters related to the DO to C ratio in respiration and submerged macrophytes decomposition (D_{SM}) but also to N recycling via the latter processes, as well as N burial and uptake (α_{NH4SM}, α_{ONSM}, β_N, $Nmax_{DW}$). Dissolved organic carbon and POC are most sensitive to parameters modulating the DO to C ratio in respiration as well as primary producer dynamics ($Resp_{DO/C}$, $growthmax_{PHY}$, $Vsettling_{PHYT}$, $Tref_{PHYT}$, $growthmax_{DW}$, m_{DW}) and submerged macrophytes decomposition. Dissolved inorganic C and TC are mainly affected by parameters modulating the growth and losses of DW, the N and C fraction released by submerged macrophytes decomposition, the CO_2 exchange with the atmosphere, β_N and β_C. Whereas the N cycle state variables (NH_4, NO_3 and TN) are highly affected by DW growth-related parameters, β_N and β_P, the P cycle state variables (PO_4, and TP) are predominantly affected by phytoplankton growth-related parameters. Finally, DO is most sensitive to submerged macrophytes decomposition and DW growth, as well as by the exchange with the atmosphere.

4. Discussion

4.1. Biogeochemical modeling of a kettle hole

We presented a model aiming to capture a number of relevant factors shaping nutrient cycling in kettle holes: DW dynamics, submerged macrophyte decomposition and water level fluctuations. Several models have been developed to gain knowledge on particular facets of DW ecology such as factors controlling DW growth (Lasfar et al., 2007), the impact of weather conditions on the onset of DW dominance (Peeters et al., 2013), the effect of population densities on DW growth rate (Driever et al., 2005) or the influence of growth inhibitors on DW populations growth (Schmitt et al., 2013). Likewise, the interplay between submerged macrophytes and water quality in shallow lakes has been assessed by means of numerical modeling (Asaeda et al., 2000; Muhammetoğlu and Soyupak, 2000). However, modeling studies of shallow systems do not consider dynamic water budgets and often opt for prespecified constant water levels (Muhammetoğlu and Soyupak, 2000; Mulderij et al., 2007; Nguyen et al., 2010). While the role of these factors has been examined individually in the literature, our multi-elemental biogeochemical model offered the opportunity to shed light on the synergistic effects of DW, submerged macrophyte decomposition and actual water level fluctuations.

The first aspect examined was the development of DW, which can potentially cover a high proportion of the water surface. Duckweed have been found to attenuate the incoming light in the water column, thereby triggering a decrease in phytoplankton biomass and/or shifts to phytoplankton community towards species adapted to poor-light conditions. Likewise, DW mats usually hamper gaseous change between water and atmosphere, favoring the establishment of anoxia and consequently an increase in nutrient release from sediments (Boedeltje et al., 2005).

The second aspect is the consideration of submerged macrophyte decay. During the study period, the submerged macrophyte *Ceratophyllum submersum* L. was present in Rittgarten. This species is typically part of the aquatic vegetation communities present in shallow eutrophic waters in agricultural landscapes of central Europe (Nagengast and Gąbka, 2017). Macrophytes can profoundly affect nutrient cycling in freshwater systems (Carpenter and Lodge, 1986). The nutrients stored during the growing season are released through decomposition upon death, thereby increasing water nutrient

Table 4
Sensitivity analysis results of the Rittgarten kettle hole biogeochemical model. Ten most influential parameters based on the mean annual relative sensitivity (MARS) of the model state variables. The percentage change applied to the calibrated parameter values (+10% or −10%) is indicated.

Chl-*a*	MARS	Duckweed	MARS	DOC	MARS	POC	MARS	DIC	MARS	TC	MARS
−10%											
Resp$_{DO/C}$	−9.99	Resp$_{DO/C}$	−9.94	Resp$_{DO/C}$	−9.85	Resp$_{DO/C}$	−9.97	Resp$_{DO/C}$	−9.77	Resp$_{DO/C}$	−9.79
β_P	−3.00	D$_{SM}$	−0.33	Toptmin	−0.91	β_P	−2.43	D$_{SM}$	−6.60	D$_{SM}$	−5.36
Ik$_{PHY}$	−1.10	α_{ONDW}	−0.25	D$_{SM}$	−0.49	D$_{SM}$	−0.82	α_{POCSM}	−2.35	α_{POCSM}	−1.88
growthmax$_{PHY}$	−1.03	α_{NH4SM}	−0.23	Vsettling	−0.38	growthmax$_{PHY}$	−0.72	α_{OPSM}	−2.25	α_{OPSM}	−1.78
D$_{SM}$	−0.91	α_{ONSM}	−0.19	KTFmin	−0.32	α_{PO4SM}	−0.51	Tmin$_{DW}$	−2.09	Tmin$_{DW}$	−1.68
Tmin$_{DW}$	1.16	growthmax$_{DW}$	0.40	Nmaxupt$_{DW}$	0.24	Tmin$_{DW}$	0.68	Tref$_{DW}$	4.00	Tref$_{DW}$	3.20
Pmax$_{PHY}$	1.22	Nmax$_{DW}$	0.58	growthmax$_{DW}$	0.32	Pmax$_{PHY}$	0.92	Nmax$_{DW}$	6.18	Nmax$_{DW}$	4.95
Tref$_{PHY}$	1.25	Tref$_{DW}$	0.90	Vbiosettling	0.55	Tref$_{PHY}$	1.08	β_C	9.67	β_C	7.69
Tref$_{DW}$	2.53	CDW$_{DW}$	1.15	β_N	0.89	Tref$_{DW}$	1.43	K$_{CO2}$	14.39	K$_{CO2}$	11.43
Vsettling$_{PHY}$	2.68	β_N	2.07	KCrefmineral	1.29	Vsettling$_{PHY}$	2.02	β_N	18.17	β_N	14.62
10%											
Chl-*a*	MARS	Duckweed	MARS	DOC	MARS	POC	MARS	DIC	MARS	TC	MARS
Resp$_{DO/C}$	−9.99	Resp$_{DO/C}$	−9.94	Resp$_{DO/C}$	−9.85	Resp$_{DO/C}$	−9.97	Resp$_{DO/C}$	−9.78	Resp$_{DO/C}$	−9.80
β_P	−3.00	β_N	−2.00	KCrefmineral	−1.04	β_P	−2.43	β_N	−7.48	β_N	−6.14
Vsettling$_{PHY}$	−1.85	m$_{DW}$	−1.55	β_N	−0.83	Vsettling$_{PHY}$	−1.41	β_C	−7.11	β_C	−5.66
Tref$_{PHY}$	−1.60	CDW$_{DW}$	−0.93	Vbiosettling	−0.55	Tref$_{PHY}$	−1.32	K$_{CO2}$	−5.36	K$_{CO2}$	−4.27
ChlaC$_{PHYT}$	−0.94	Nmax$_{DW}$	−0.45	Nmaxupt$_{DW}$	−0.18	Pmax$_{PHY}$	−0.70	Nmax$_{DW}$	−5.18	Nmax$_{DW}$	−4.15
α_{PO4SM}	0.58	α_{NH4SM}	0.22	KTFmin	0.33	α_{PO4SM}	0.51	α_{POCSM}	2.06	α_{POCSM}	1.65
D$_{SM}$	0.90	α_{ONDW}	0.26	Tref$_{DW}$	0.33	D$_{SM}$	0.81	Tmin$_{DW}$	3.20	Tmin$_{DW}$	2.57
growthmax$_{PHY}$	1.25	D$_{SM}$	0.27	m$_{DW}$	0.43	growthmax$_{PHY}$	0.90	Tref$_{DW}$	3.21	Tref$_{DW}$	2.61
growthmax$_{DW}$	1.66	growthmax$_{DW}$	0.42	D$_{SM}$	0.51	growthmax$_{DW}$	0.95	D$_{SM}$	6.18	D$_{SM}$	5.03
Ik$_{PHY}$	2.30	Tref$_{DW}$	0.46	Toptmin	1.20	Ik$_{PHY}$	1.32	m$_{DW}$	7.71	m$_{DW}$	6.21
−10%											
NH$_4$	MARS	NO$_3$	MARS	TN	MARS	TP	MARS	PO$_4$	MARS	DO	MARS
growthmax$_{DW}$	15.46	growthmax$_{DW}$	21.67	Resp$_{DO/C}$	−9.21	Resp$_{DO/C}$	−9.94	Ik$_{PHY}$	24.47	Tref$_{DW}$	43.93
β_N	4.72	Toptnit$_{sed}$	1.14	Ik$_{PHY}$	−1.13	β_P	−1.45	growthmax$_{PHY}$	10.09	Tmin$_{DW}$	29.16
Nmaxupt$_{DW}$	2.46	Vsettling$_{PHY}$	1.08	growthmax$_{PHY}$	−0.72	Tref$_{DW}$	−1.00	Pmaxupt$_{PHY}$	5.22	Vsettling$_{PHY}$	19.18
CDW$_{DW}$	0.95	D$_{SM}$	1.05	β_P	−0.68	D$_{SM}$	−0.70	β_P	4.46	Pmax$_{PHY}$	10.78
K$_{DWden}$	0.94	Pmax$_{PHY}$	1.02	Tref$_{DW}$	−0.47	Tref$_{PHY}$	−0.64	growthmax$_{DW}$	4.32	I$_{sat}$	9.34
I$_{sat}$	−0.52	β_C	−2.39	K$_{DWden}$	0.59	NmaxuptDW	0.27	Mphy	−1.11	β_P	−0.95
Tmin$_{DW}$	−0.58	β_P	−3.01	Vsettling$_{PHY}$	0.68	growthmax$_{DW}$	0.69	Pmax$_{PHY}$	−1.15	β_C	−0.98
Ik$_{PHY}$	−0.83	growthmax$_{PHY}$	−3.08	NmaxuptDW	1.17	Pmaxupt$_{PHY}$	0.82	Tref$_{PHY}$	−1.85	Resp$_{DO/C}$	−9.69
Tref$_{DW}$	−1.09	Ik$_{PHY}$	−4.56	β_N	3.27	growthmax$_{PHY}$	1.11	Vsettling$_{PHY}$	−2.08	β_N	−18.52
Resp$_{DO/C}$	−9.11	Resp$_{DO/C}$	−9.52	growthmax$_{DW}$	6.22	Ik$_{PHY}$	1.83	Resp$_{DO/C}$	−9.94	K$_{O2}$	−31.17
10%											
NH$_4$	MARS	NO$_3$	MARS	TN	MARS	TP	MARS	PO$_4$	MARS	DO	MARS
Resp$_{DO/C}$	−9.11	Resp$_{DO/C}$	−9.52	Resp$_{DO/C}$	−9.21	Resp$_{DO/C}$	−9.94	Resp$_{DO/C}$	−9.94	m$_{DW}$	−9.87
β_N	−2.88	Vsettling$_{PHY}$	−3.16	β_N	−2.25	β_P	−1.45	growthmax$_{PHY}$	−1.36	Resp$_{DO/C}$	−9.71
growthmax$_{DW}$	−0.81	β_P	−3.01	Vsettling$_{PHY}$	−0.84	growthmax$_{DW}$	−0.77	Pmaxupt$_{PHY}$	−1.32	Tmin$_{DW}$	−1.31
Nmaxupt$_{DW}$	−0.54	Tref$_{PHY}$	−2.39	β_P	−0.68	Ik$_{PHY}$	−0.76	Ik$_{PHY}$	−0.87	β_P	−0.95
CDW$_{DW}$	−0.47	Pmax$_{PHY}$	−2.11	Tref$_{PHY}$	−0.50	growthmax$_{PHY}$	−0.61	growthmax$_{DW}$	−0.69	D$_{SM}$	−0.91
α_{ONDW}	0.54	Pmaxupt$_{PHY}$	1.09	growthmax$_{PHY}$	0.45	D$_{SM}$	0.72	β_P	4.46	growthmax$_{PHY}$	14.96
I$_{sat}$	0.95	growthmax$_{PHY}$	1.11	Ik$_{PHY}$	0.56	Tmin$_{DW}$	0.78	Tmin$_{DW}$	7.68	K$_{O2}$	19.34
Tmin$_{DW}$	4.64	Ik$_{PHY}$	1.30	Tmin$_{DW}$	2.09	Vsettling$_{PHY}$	0.81	Tref$_{PHY}$	8.87	β_N	25.87
Tref$_{DW}$	19.21	Tref$_{DW}$	29.92	Tref$_{DW}$	7.51	m$_{DW}$	1.19	m$_{DW}$	11.93	Ik$_{PHY}$	26.59
m$_{DW}$	20.39	m$_{DW}$	30.96	m$_{DW}$	8.07	Tref$_{PHY}$	1.58	Vsettling$_{PHY}$	12.93	growthmax$_{DW}$	32.97

concentrations (Shilla et al., 2006 and references therein). In addition, microbial decomposition of dead plant tissues entails DO consumption which in turn results in decreased DO concentrations (Godshalk and Wetzel, 1978). In particular, submerged macrophyte can be expected to impact nutrient seasonal trends since they are typically degraded within one year upon death, which is in part due to the smaller amount of lignified tissues in comparison to emergent macrophytes, rendering them more readily degradable by microorganisms (Kirschner et al., 2001). To accommodate the broader role of macrophyte decomposition on the kettle hole biogeochemistry, a relevant term was added to both the dissolved and particulate nutrient equations.

The third relevant aspect for a small system, like the studied kettle hole, that is accounted for by the model is the dynamic water volume. Water level fluctuations affect the biota and physical environment of freshwater ecosystems; for instance, changes in light penetration might change the primary production proportion between pelagic and littoral zone (Leira and Cantonati, 2008). Water level fluctuations might lead to shifts between clear-water and turbid state, nutrient concentrations or variations in settling fluxes (Coops et al., 2003; Williams 2006; Kleeberg et al., 2016a). Similarly, in prairie potholes in North America, water level fluctuations have been reported to cause changes in the relative abundance of macrophytes species, when water level variations were within the 0.5 m during wet-dry cycles, and changes in species composition (i.e. disappearance of emergent species), when water level varied between 1.5 and 2 m (Van der Valk, 2005).

4.2. Seasonal dynamics of model state variables

During the study period, the chemical and biological components simulated by the model displayed pronounced seasonal variations (Fig. 2). Phytoplankton was characterized by a biomass maximum in summer and moderate to low values during the rest of the year. Duckweed reached the maximum during summer and decreased in autumn, being absent during winter and most of spring. The concentration of POC showed a threefold increase between winter and spring, decreasing towards summer. Concomitant to minimum DO concentrations ($\approx$1.5 mg L^{-1}), DOC, NH$_4$, and PO$_4$ increased noticeably in July 2014. It is also worth noting that Rittgarten underwent a significant water level decrease from summer 2013 onwards, falling dry in August 2014 (Kleeberg et al., 2016a). However, there was no distinct increase pattern of nutrients concentration during the same period, which suggests that water level decrease was not the main driver of the observed nutrient seasonal dynamics. Chlorophyll-*a* and nutrient concentrations observed in Rittgarten lay within the typically reported ranges for highly eutrophic shallow systems (Wetzel, 2001). The current trophic state could also be associated with the historical sediment burdens associated with agricultural intensification in its catchment from 1853 onwards (Kleeberg et al., 2016b). The actual magnitude of the TOC settled on the bottom of Rittgarten, as well as the apparent causal connection between Chl-*a* peaks and TOC settling fluxes are reflective of the high phytoplankton primary productivity sustained by the system (Kleeberg et al., 2016a).

The temporal dynamics of the two primary producers and their seasonal succession in the kettle hole were reasonably predicted by the model (Fig. 2). During the study period, DW developed between late spring and early autumn in Rittgarten, peaking in summer, which can be considered as the typical pattern in temperate shallow lakes (Sayer et al., 2010). In lakes where free-floating macrophytes typically prevail, situations of complete absence and dense cover alternate as a function of lake size, morphometry, water level, nutrient concentrations and temperature (O'Farell et al., 2009). The development of dense DW mats in shallow waters has often been linked to a strong decrease in phytoplankton biomass due to severe light attenuation in the water column (Abdel-Tawwab, 2006; Pasztaleniec et al., 2013). Interestingly, there was an extended period (summer to early autumn 2013) in Rittgarten, when relatively high phytoplankton biomass (Chl-*a* up to $80\ \mu g\ L^{-1}$) was found under a dense DW cover. This cover likely favored phytoplankton groups well adapted to low-light availability. Poor-light climate under free-floating macrophyte cover typically triggers changes in the composition of the algal assemblage, with a higher contribution of groups such as cyanophytes, diatoms, cryptophytes, and euglenoids, well adapted to low light conditions (de Tezanos Pinto and O'Farrell, 2014). Similar to Rittgarten observations, intermediate to high Chl-*a* concentrations ($20–100\ \mu g\ L^{-1}$) were found in different shallow freshwater systems under dense free-floating macrophyte mats, and this was attributed to differential efficiency for N and P uptake by phytoplankton and floating macrophytes (de Tezanos Pinto and O'Farrell, 2014). Based on the predicted cumulative annual fluxes for nutrient uptake (Fig. 3), we hypothesize that competition for nutrients explains the relatively high phytoplankton biomass in the presence of DW in Rittgarten. Whereas phytoplankton utilized most of the P consumed, duckweed removed most N. The potential of phytoplankton to reduce DW growth by P depletion has been pinpointed by Szabó et al. (1999). Similar to the latter study, we also found that duckweed internal P concentrations in Rittgarten (2 mg P g dry weight^{-1}, Kleeberg et al., 2016a) were below the optimal growth range for DW species they reported (5.7–26.6 mg P g dry weight^{-1}).

4.3. Internal nutrient recycling

According to our model predictions, the nutrients made available via mineralization and phytoplankton basal metabolism are not sufficient to cover the free floating macrophytes and phytoplankton P and N requirements. This indicates that other internal nutrient recycling processes such as submerged macrophyte decomposition and nutrient release from sediments might account for a considerable proportion of the bioavailable nutrient concentrations. Even though our model predicts an approximate equilibrium between P sources and sinks at the Rittgarten sediments over the course of a year, we also found periods of high P release in summer that likely fuel the growth of primary producers. In summer, PO_4 increased noticeably concomitant to minimum DO values (≈ 1.5 mg L^{-1}), pointing to enhanced diffusive flux of PO_4 upon release at the sediment-water interface due to the reduction of iron-bound P under hypoxic conditions. The mobilization of P to the sediment dissolved fraction and the subsequent upward transport of the dissolved species is one of the major P release pathways (Boström et al., 1988). In fact, in a study comprising $\approx$80 kettle holes in Northeastern Germany, internal P release during hypoxia explained as much as two-thirds of both temporal and spatial variability of physical and chemical parameters (Lischeid and Kalettka, 2012). An additional factor that might favor P release in Rittgarten is the formation of iron sulphides by reducing the amount of iron available to bind P (Kleeberg et al., 2016a). In Rittgarten, not only PO_4 but also NH_4 augmented during summer parallel to low DO. A possible explanation is the accumulation of NH_4 in pore water of anoxic sediments as a result of inhibition of bacterial nitrification and low NH_4 microbial assimilation rates, followed by NH_4 released to the water column (Beutel, 2006). The latter authors stressed that whereas the release of PO_4 and iron takes place when oxidation-reduction potential falls below $\approx$200 mV, the release of NH_4 starts when anoxia onsets. In Rittgarten, the oxidation-reduction potential averaged 122 mV in summer. Lastly, sediments in kettle holes are subjected to significant temperature seasonal variations, thereby reaching high temperatures in summer, which in turn accelerate mineralization rates and nutrient sediment release (Meerhoff and Jeppesen, 2009).

In addition to nutrient sediment release, the decay of submerged macrophytes to PO_4 and NH_4 availability seems to play an important role in sustaining the high phytoplankton and DW biomass observed in Rittgarten. Our model predicted a predominant contribution of submerged macrophyte decomposition to N and P concentrations at the annual scale; namely, 36% and 17% of the P and N required by phytoplankton and DW. The role of submerged macrophytes decay in supporting phytoplankton growth has been stressed in previous studies (Kim et al., 2013). An increase in phytoplankton biomass following the release of easily bioavailable P by senescent submerged *Myriophyllum spicatum* L. was reported in enclosures placed in a shallow reservoir (Landers, 1982). Likewise, late-summer phytoplankton blooms have been observed in a number of small shallow temperate lakes following plant senescence (Sayer et al., 2010).

The fact that TN measured concentrations were underestimated by the model might be explained by the use of fixed parameters values for submerged macrophyte stoichiometry. We hypothesize that the implementation of internal storage capacity levels for nutrient uptake, modulated by both intracellular and extracellular nutrient concentrations, would enable a more dynamic and realistic approximation to the nutrients contained in decaying tissues that would result in a better model fit to the observations. Further experimental studies regarding the decomposition dynamics of macrophytes in Rittgarten are necessary to better portray their role on the nutrient cycling of the system.

4.4. Allochthonous inputs

Allochthonous inputs, not considered by the model, could be partially responsible for the underestimation of the simulated TN and POC concentrations. Empirical results from Rittgarten and nearby kettle holes are suggestive of a predominant contribution of autochthonous sources to C budget (Attermeyer et al., 2017). The organic matter supplied by primary producers likely controls bacterial production and biogeochemical processes in these water bodies. Further, coarse materials likely originated from macrophytes were the main constituent of sediment traps material in Rittgarten (Kleeberg et al., 2016a). However, input from plant debris other than the dominating aquatic plants, ambient surface soils, as well as crops from the catchment were also detected. The latter finding suggests that the underestimation of nutrient concentrations by the model might partially stem from the omission of external nutrient inputs.

Neither significant nutrient input via particulate atmospheric deposition, nor surface run-off associated to snowmelt or precipitation occurred in Rittgarten during the study period. Two dust deposition samplers (Hoffmann et al., 2008) were placed at the shore of the kettle hole, providing evidence of minimal atmospheric input by wind erosion during the study period. Based on the annual atmospheric N deposition of 12 kg ha^{-1} modeled for the region (Map server of the German Environment Agency, Kruit et al. 2014), the average atmospheric N input to the kettle hole is 0.6 kg y^{-1}, which represents a small fraction of the simulated N fluxes (Fig. 3). Hence, particulate nutrient inputs via atmospheric deposition were ruled out. Nutrient inputs and losses via groundwater and interflow were similarly considered negligible given that: i) the mean water residence time within the kettle hole basins located in the Uckermark region was in the order of ≈5 y^{-1} for the 2013–2015 period (Nitzsche et al., 2017), which is very long compared to the kinetics of internal processes; ii) hydrological connectivity and conductivity of glacial tills is low, and therefore mass transport in these systems is assumed to be biogeochemically, rather than hydrologically, controlled (Reverey et al., 2016); iii) the impact of groundwater on water quality in 80 kettle holes in Northeast Germany was found to be minor (Lischeid and Kalettka, 2012); and iv) the groundwater quality can be regarded as stable during the studied time scale (Lischeid et al., 2017). However, the contribution of external inputs during other time periods or longer time spans cannot be ruled out. Indeed, evidence from kettle hole sediments focusing on long time spans identified significant inputs of eroded soil, nutrients and organic matter associated with the long-term intensive agricultural practices in kettle hole catchments (Frielinghaus and Vahrson, 1998; Kleeberg et al., 2016b; Nitzsche et al., 2017).

4.5. Kettle holes as sources of carbon

The role of ponds in global C cycling has been recently acknowledged based on their high areal extent and intensity of C processing (Downing, 2010). Evidence suggests that the magnitude of areal CO_2 efflux rates from small ponds might be comparable to those of large lakes or reservoirs but also that areal C burial rates of ponds practically surpass those of any other ecosystem (Torgersen and Branco, 2008; Downing, 2010). However, the limited number of studies regarding CO_2 and CH_4 fluxes in temperate small (<1 ha) shallow ponds still hinders the understanding of inland waters C cycling (Holgerson, 2015).

The developed model allows to quantify the nutrient fluxes associated to the simulated biogeochemical processes, thereby providing insights into the nutrient budgets of Rittgarten and those kettle holes structurally and functionally similar (i.e., comparable geomorphology, hydroperiod and dominant aquatic vegetation). According to our model predictions, the amount of C buried into deeper sediment at the annual scale (26.05 kg y^{-1}) was lower than the amount of C released to the atmosphere (45.35 kg y^{-1}). In particular, the predicted CO_2 loss to the atmosphere of ≈45 g C m^{-2} y^{-1} lays within the range reported for different ponds and inland waters <100 ha (Torgersen and Branco, 2008; Premke et al., 2016). This result is in agreement with the projections that small water bodies in the moraine landscape of NE Germany release CO_2 on an annual scale (Premke et al., 2016). Despite their small size, covering around 10% of the studied area, their inclusion in the calculations of C landscape budgets resulted in a shift from C removal to net C emission, pinpointing their role as C hotspots in the landscape.

4.6. Wet-dry cycles and water level fluctuations

Information on how the intermittent nature of kettle holes affects the seasonal dynamics of our model state variables and the magnitude of the associated nutrient fluxes is scarce. The majority of kettle holes experience wet-dry cycles, and the associated drying and rewetting events represent biogeochemically hot moments. More specifically, short desiccation periods can lead to a decrease in the sediment organic matter content as a result of aerobic decomposition or photodegradation in exposed sediments, which might prevent C and nutrient sequestration in kettle hole sediments as well as net gain in sediment depth (Williams et al., 2001; Reverey et al., 2016). Rewetting events have been linked to enhanced CO_2 efflux due to increased microbial activity stemming from high C lability upon rewetting (Gilbert et al., 2017). Conversely, desiccation periods can facilitate the encroachment by terrestrial plants (Nitzsche et al., 2017), that would contribute to sediment C sequestration upon death at rising water levels. In temporary ponds, C and N losses via mineralization and nitrification/denitrification and subsequent outgassing were observed during rewetting, whereas P bioavailability was found to increase via enhanced mineralization (Reverey et al., 2016). It is notable that the magnitude and duration of preceding drought can regulate the mentioned nutrient fluxes. Also, when the sediments are exposed to the air, inorganic sulphur accumulated into inundated kettle hole sediments via external sulphate input oxidizes forming sulfuric acid that upon kettle hole rewetting leads to a temporary water pH decrease (Thomas Kalettka, personal communication). In the present modeling exercise, we provided insights into C, N and P annual budgets of Rittgarten based on a wet phase. However, the consideration of the latter processes is essential to quantify elemental annual budgets in kettle holes over time spans comprising wet-dry cycles.

The effects of water level fluctuations within the wet phase on the prevalence of different primary producers groups and their annual succession in kettle holes is currently unknown. Changes in water level have been proposed to drive shifts between free-floating macrophytes, submerged macrophytes and phytoplankton regimes (de Tezanos Pinto and O'Farrell, 2014). In the shallow lake Võrtsjärv, low water levels resulted in improved underwater light climate, causing the collapse of *Limnothrix* species and favoring the growth of the synechococcal *Cyanonephron styloides* Hickel and the submerged macrophyte *Myriophyllum spicatum* Linnaeus (Nõges and Nõges, 1999). In the shallow urban wetland Cootes Paradise Marsh, high and low water levels apparently lead to light limitation and too dry conditions for optimal germination of submerged aquatic vegetation, respectively (Chow-Fraser, 2005). Most available evidence draws from studies comprising long time spans (several years). The understanding of aquatic vegetation seasonal dynamics in small temperate shallow lakes is still limited (Sayer et al., 2010).

5. Conclusions

We developed a process-based biogeochemical model able to reproduce the seasonal dynamics of nutrients (albeit those of the nitrogen cycle to a lesser degree), phytoplankton and duckweed observed in Rittgarten. The model outputs were most sensitive to the DO to C ratio in respiration, submerged macrophyte decomposition, the fraction of N and P buried into deep sediment and parameters related to growth and losses of primary producers. Our results suggest that the establishment of a phytoplankton community well adapted to low light availability, together with the differential use of N and P by DW and phytoplankton can explain the coexistence of both primary producers groups in Rittgarten. The nutrient budget predictions suggest that the release of nutrients from sediments and via submerged macrophyte decomposition are the major processes sustaining the growth of phytoplankton and DW in the system. Our results indicated that Rittgarten releases CO_2 to the atmosphere at the annual scale, and by extension, we expect that so do kettle holes structurally and functionally similar. However, in order to achieve a comprehensive understanding of kettle hole biogeochemical functioning, further empirical studies are necessary with regard to i) the effects of water level fluctuations on the seasonal dynamics of chemical and biological components in the kettle hole; ii) the drivers of primary producers prevalence and succession; and iii) the relative contribution of external sources and internal nutrient recycling mechanisms to bioavailable nutrient concentrations.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2018.04.014.

Acknowledgements

We would like to thank Mathias Hoffmann for providing PAR data from CarboZALF experimental area, Carsten Hoffmann for atmospheric deposition data, and Christoph Merz for hydrological information. We are grateful to Steffen Gliege, Florian Revery and Carlos Acame for valuable comments during manuscript development. The study was performed within the LandScales project and funded by the Pact for Innovation and Research of the Gottfried Wilhelm Leibniz Scientific Community. We also thank two anonymous reviewers for their helpful comments on a previous version of the manuscript.

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